

Clinical Risk Assessment Report: Patient 72

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 14.3% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.65x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Large Zone Emphasis (CT) (GLZLM_LGZE.CT), which reduced the patient's absolute risk by 1.8%. Biologically, this feature reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: 0.44 [Avg (69th %ile)] (+1.7%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.
- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 0.5 [Bottom 10%] (+1.4%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: -0.19 [Avg (52th %ile)] (-1.8%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)

Value: -0.39 [Avg (42th %ile)] (-1.3%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.
- Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT)

Value: 1.12 [Top 14%] (-1.2%)

Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.
- Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT)

Value: -0.13 [Avg (54th %ile)] (-0.8%)

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.
- 75th Percentile Tumor Density (CONVENTIONAL_HUQ3)

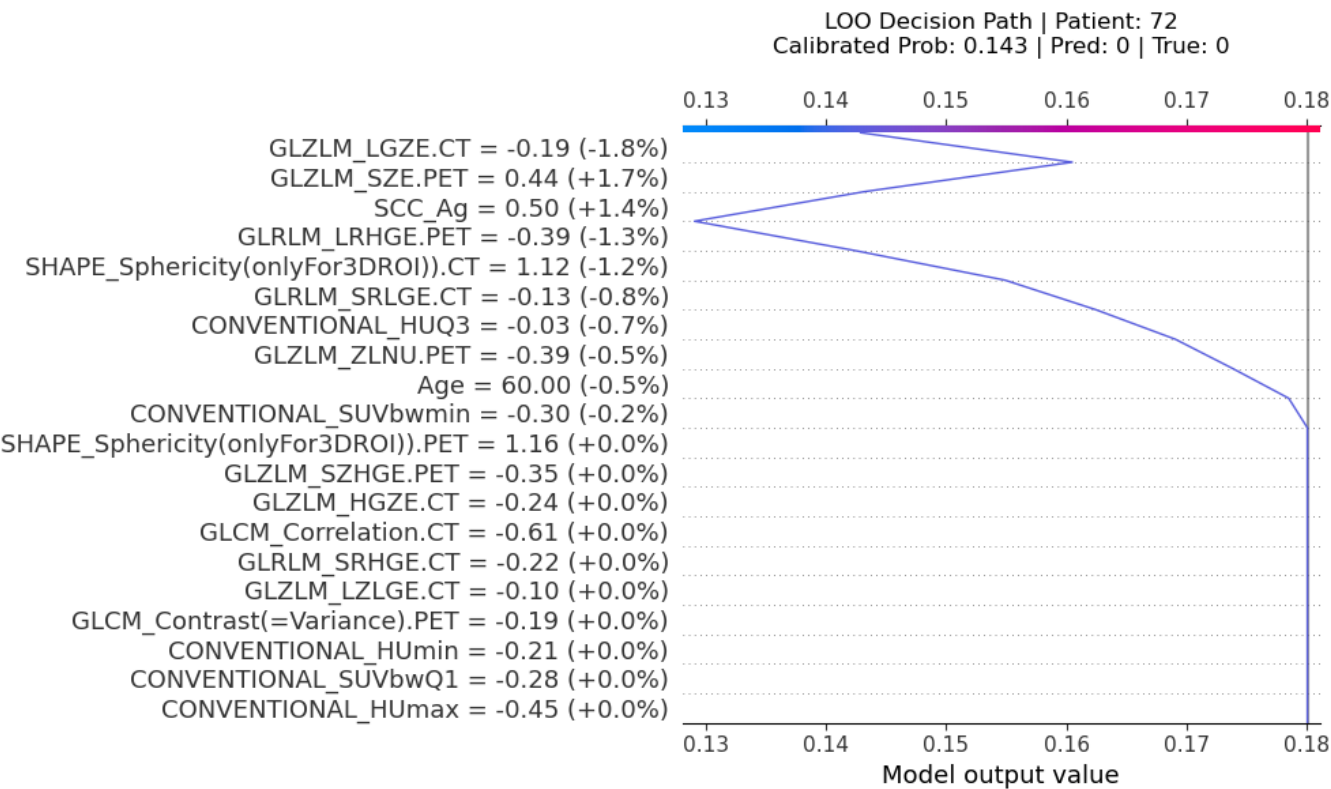
Value: -0.03 [Avg (36th %ile)] (-0.7%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.
- PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET)

Value: -0.39 [Bottom <1%] (-0.5%)

Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.

